

HW4code.R

Sean's Desktop

2026-05-14

```
#Remove previous data
rm(list=ls())
#setwd("C:/Users/Sean's Desktop/Documents/GitHub/Math-447/S26HW4")

#Thanks to Nolan Mickenham for suggesting the Dinosaur Genera Dataset.
#https://www.kaggle.com/datasets/canozensoy/dinosaur-genera-dataset

#Be sure to do Session -> Set Working Directory -> To Source File Location
dino_data <- read.csv("dinoDatasetCSV.csv", header=TRUE)
dino_data2 <- dino_data[which(dino_data$diet %in% c("Carnivore","Herbivore")),c(4,5,6,7)]

dino_data2$log_len_wt <- log(dino_data2$length_m/dino_data2$weight_kg)
dino_data2$log_len_ht <- log(dino_data2$length_m/dino_data2$height_m)

dino_data2$carnivore <- as.integer(dino_data2$diet == "Carnivore")

rowNAs <- which(is.na(rowSums(dino_data2[, -1])))
dino_data2 <- dino_data2[-rowNAs, -1] #removing carnivore vs. herbivore
columns <- colnames(dino_data2)

#Glossary
#Predictors
#length_m: Length in meters
#weight_kg: weight_kgin kilograms
#height_m: Height in meters
#log_len_wt: Log(Length/Weight)
#log_len_ht: Log(Length/Height)
#Response
#carnivore: 1 if carnivore, 0 if herbivore

#standardizing each predictor variable
dino_data2_x <- data.frame(scale(as.matrix(dino_data2[, -6])))
dino_data_sd <- cbind(dino_data2[, 6], dino_data2_x)
colnames(dino_data_sd) <- columns[c(6, 1:5)]

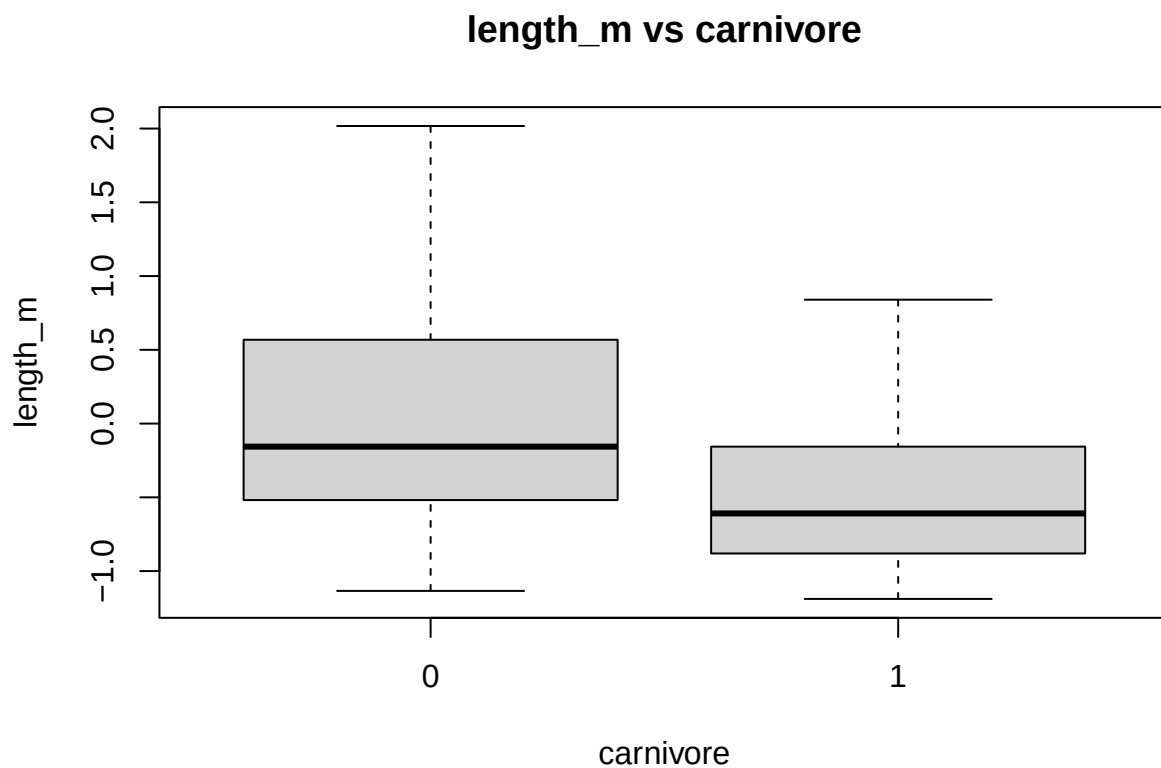
#For the homework, run the following.
#If you do not have MASS or class, install them first using install.packages().
library(MASS)
library(class)

#For the example dataset dino_data_sd, the following is necessary.
library(ISLR)
```

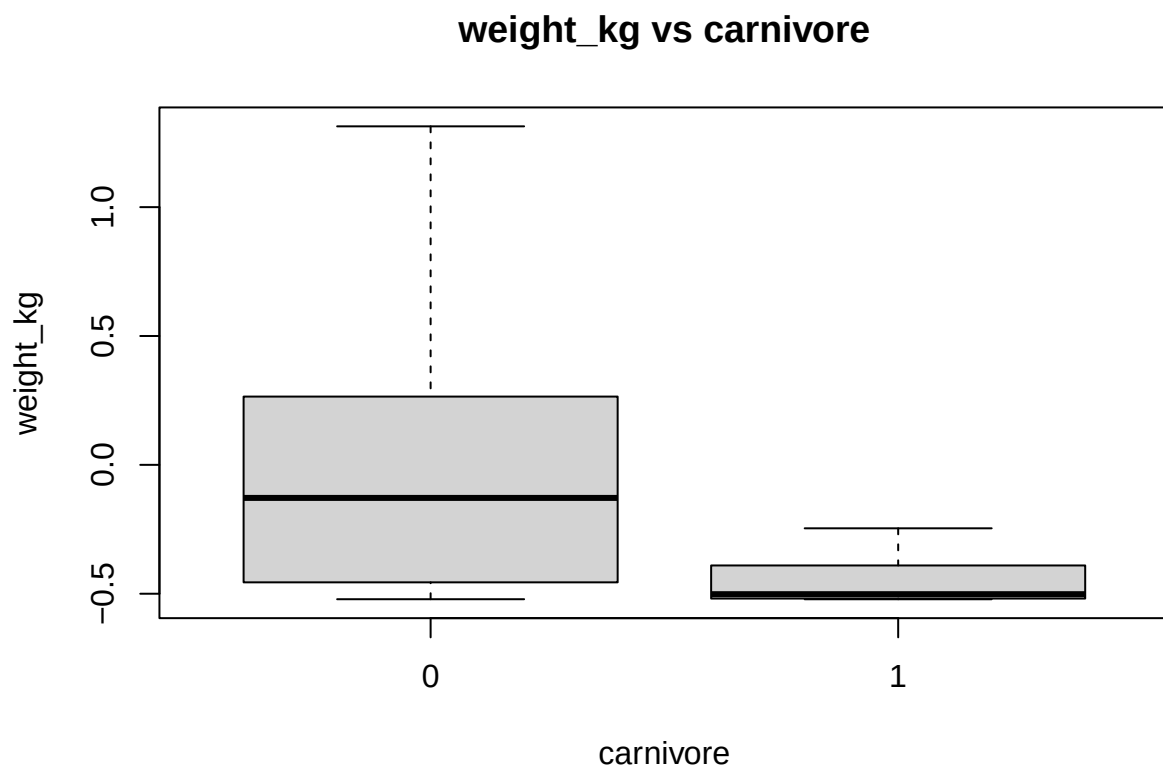
```
## Warning: package 'ISLR' was built under R version 4.5.3
#Note: The following line is necessary only for the dino_data_sd data.
attach(dino_data_sd)

#R tip: For large data, use head(dino_data_sd).
#This displays the header and the first six rows of the data.

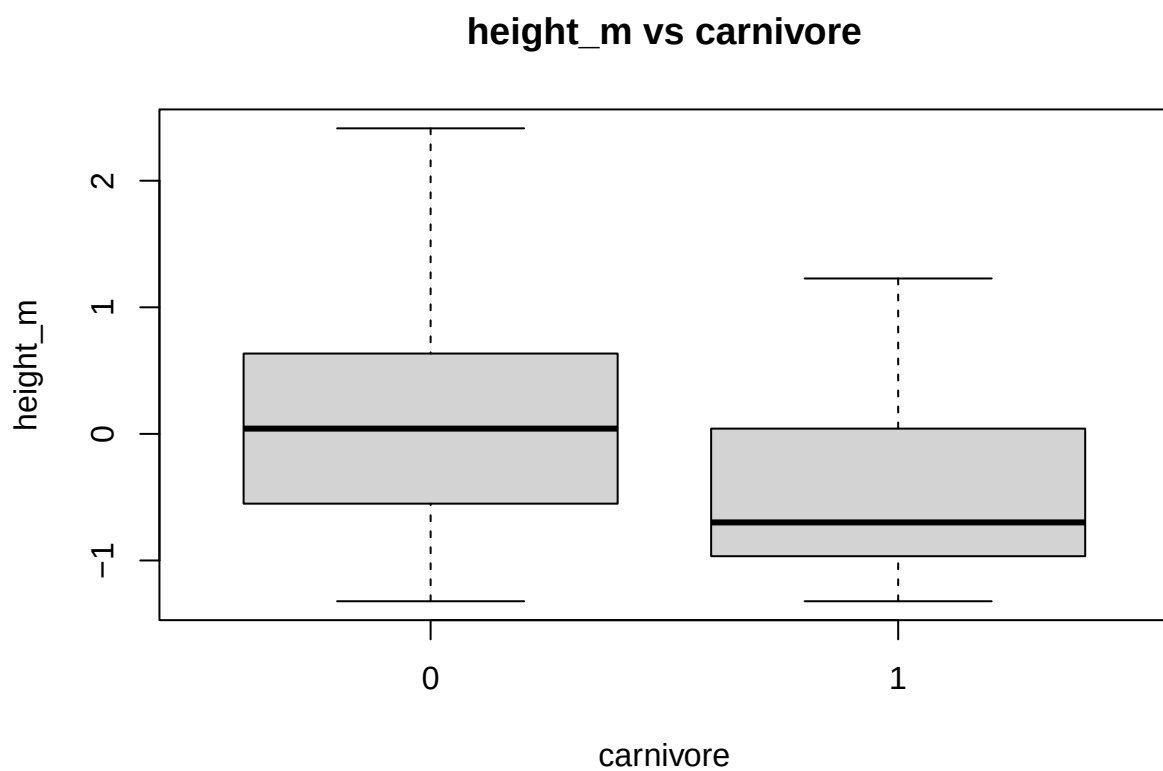
#b
#Create a boxplot for each predictor against the response.
#In the example below, length_m is a predictor and carnivore is the response.
#note that outline=FALSE is used to remove any outliers from the display.
boxplot(length_m ~ carnivore, data = dino_data_sd, main = "length_m vs carnivore", outline=FALSE)
```



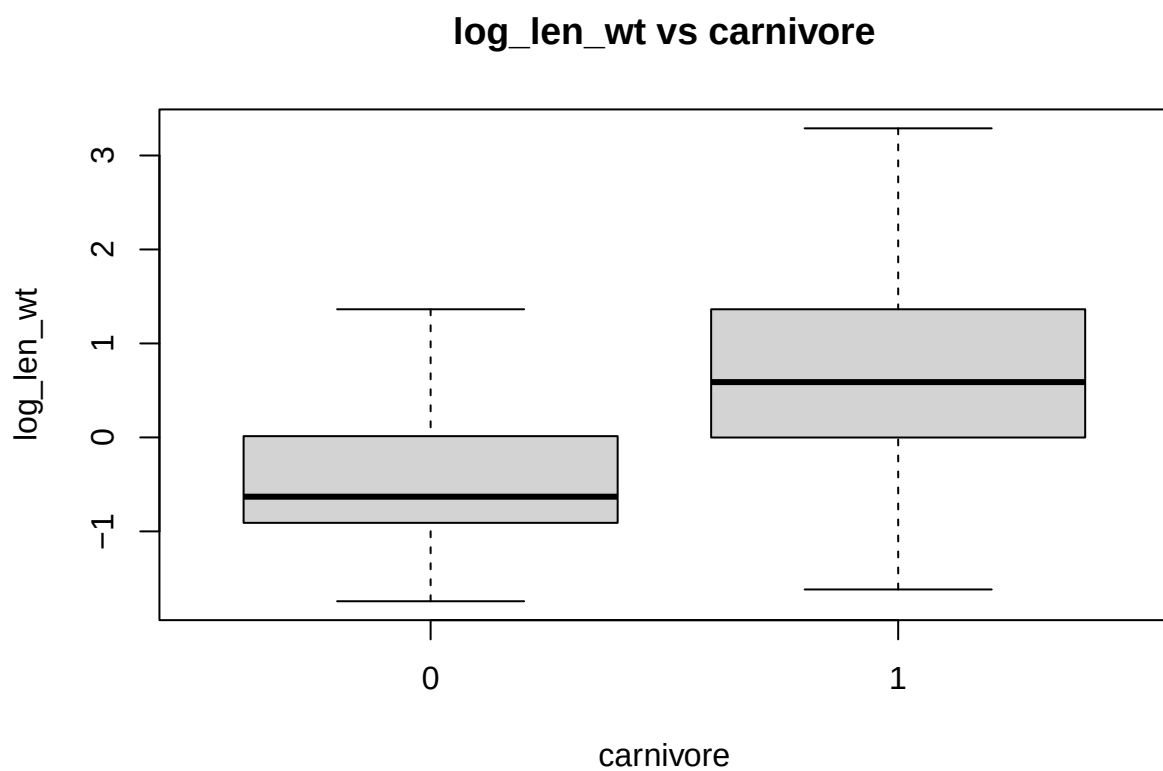
```
boxplot(weight_kg ~ carnivore, data = dino_data_sd, main = "weight_kg vs carnivore", outline=FALSE)
```



```
boxplot(height_m ~ carnivore, data = dino_data_sd, main = "height_m vs carnivore", outline=FALSE)
```

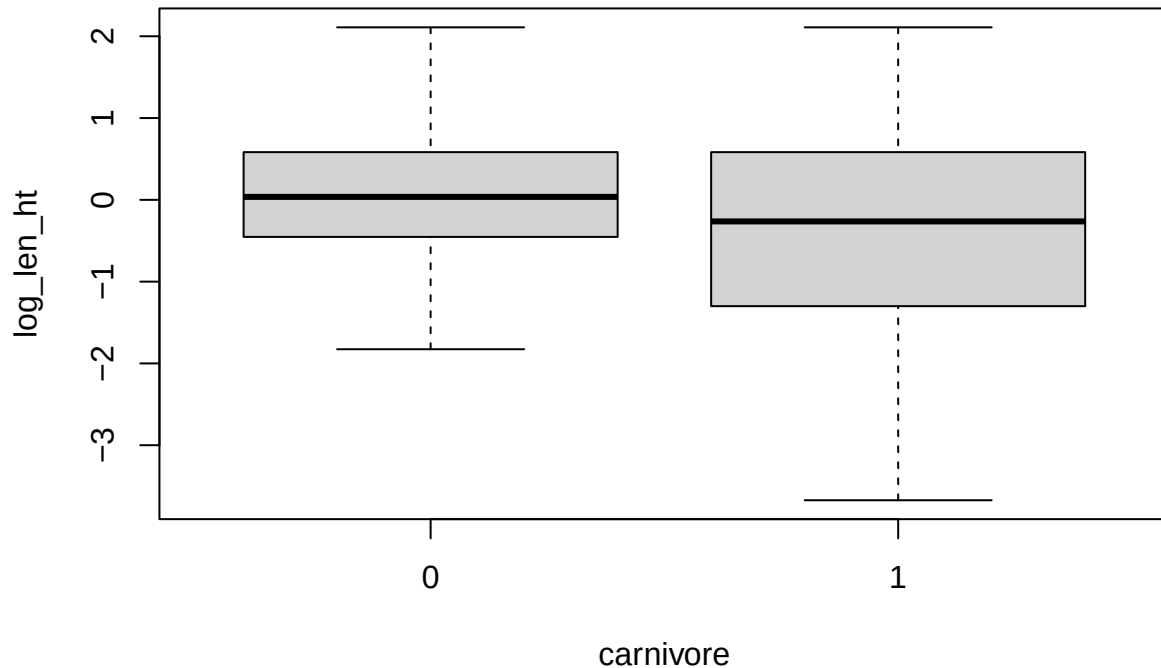


```
boxplot(log_len_wt ~ carnivore, data = dino_data_sd, main = "log_len_wt vs carnivore", outline=FALSE)
```



```
boxplot(log_len_ht ~ carnivore, data = dino_data_sd, main = "log_len_ht vs carnivore", outline=FALSE)
```

log_len_ht vs carnivore



```
#c
set.seed(11, sample.kind="Rejection") #Use this seed for your homework
train <- sample(1:dim(dino_data_sd)[1], dim(dino_data_sd)[1]/2)
test <- -train
dino_data_sd.train <- dino_data_sd[train, ]
dino_data_sd.test <- dino_data_sd[test, ]
carnivore.test <- carnivore[test]

#d
fit.lda <- lda(carnivore ~ length_m + weight_kg + height_m + log_len_wt + log_len_ht, data = dino_data_sd, subset = train)
fit.lda

## Call:
## lda(carnivore ~ length_m + weight_kg + height_m + log_len_wt +
##      log_len_ht, data = dino_data_sd, subset = train)
##
## Prior probabilities of groups:
##      0      1
## 0.667129 0.332871
##
## Group means:
##      length_m weight_kg height_m log_len_wt log_len_ht
## 0  0.1919345 0.2141241 0.2090631 -0.3676241 0.09805966
## 1 -0.4236331 -0.4078841 -0.4462007 0.6581345 -0.18188271
##
## Coefficients of linear discriminants:
##              LD1
```

```

## length_m    -0.3060132
## weight_kg   -0.5584939
## height_m    1.1812672
## log_len_wt  1.5273154
## log_len_ht  0.1538712

pred.lda <- predict(fit.lda, dino_data_sd.test)
#Confusion matrix
table(pred.lda$class, carnivore.test)

##      carnivore.test
##           0      1
##    0 416 117
##    1   50 139

#Test error rates
mean(pred.lda$class != carnivore.test)

## [1] 0.2313019

#what is test error of model?
#Discuss if predicted predictors were accurate

#e
#Simply use all the predictor variables for your homework question.
fit.qda <- qda(carnivore ~ length_m + weight_kg+ height_m + log_len_wt + log_len_ht, data = dino_data_sd.test)
fit.qda

## Call:
## qda(carnivore ~ length_m + weight_kg + height_m + log_len_wt +
##      log_len_ht, data = dino_data_sd, subset = train)
##
## Prior probabilities of groups:
##           0           1
## 0.667129 0.332871
##
## Group means:
##      length_m weight_kg height_m log_len_wt log_len_ht
## 0  0.1919345  0.2141241  0.2090631 -0.3676241  0.09805966
## 1 -0.4236331 -0.4078841 -0.4462007  0.6581345 -0.18188271

pred.qda <- predict(fit.qda, dino_data_sd.test)
table(pred.qda$class, carnivore.test)

##      carnivore.test
##           0      1
##    0 236  27
##    1 230 229

mean(pred.qda$class != carnivore.test)

## [1] 0.3559557

#f
#Simply use all the predictor variables for your homework question.
fit.glm <- glm(carnivore ~ length_m + weight_kg+ height_m + log_len_wt + log_len_ht,
               data = dino_data_sd, family = binomial, subset = train)

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```

```
fit.glm

##
## Call:  glm(formula = carnivore ~ length_m + weight_kg + height_m + log_len_wt +
##        log_len_ht, family = binomial, data = dino_data_sd, subset = train)
##
## Coefficients:
## (Intercept)      length_m      weight_kg      height_m      log_len_wt      log_len_ht
##      -1.9772      -0.4360      -5.8118       3.2509       1.7246       0.1757
##
## Degrees of Freedom: 720 Total (i.e. Null);  715 Residual
## Null Deviance:      917.4
## Residual Deviance: 640.2      AIC: 652.2
```

```
summary(fit.glm)

##
## Call:
## glm(formula = carnivore ~ length_m + weight_kg + height_m + log_len_wt +
##      log_len_ht, family = binomial, data = dino_data_sd, subset = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.9772     0.2671  -7.404 1.33e-13 ***
## length_m     -0.4360     0.8857  -0.492 0.622504
## weight_kg    -5.8118     0.9311  -6.242 4.33e-10 ***
## height_m      3.2509     0.8512   3.819 0.000134 ***
## log_len_wt     1.7246     0.2178   7.917 2.43e-15 ***
## log_len_ht     0.1757     0.1774   0.990 0.321993
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 917.39  on 720  degrees of freedom
## Residual deviance: 640.24  on 715  degrees of freedom
## AIC: 652.24
##
## Number of Fisher Scoring iterations: 8

probs <- predict(fit.glm, dino_data_sd.test, type = "response")
pred.glm <- rep(0, length(probs))
pred.glm[probs > 0.5] <- 1
table(pred.glm, carnivore.test)

##           carnivore.test
## pred.glm    0    1
##           0 394  68
##           1  72 188

mean(pred.glm != carnivore.test)

## [1] 0.1939058
```

```
#g
#For this homework, create a plot of the misclassification error rate vs. K
```



```

#for K = 1, 2, 3, ..., 10.
#Based on the plot, choose the 'best' K.
#Optionally, compute sensitivity and specificity for different values of K (not coded).
train.X <- cbind(length_m, weight_kg, height_m, log_len_wt, log_len_ht)[train, ]
test.X <- cbind(length_m, weight_kg, height_m, log_len_wt, log_len_ht)[test, ]
train.carnivore <- carnivore[train]

error <- 1:10

for (i in 1:10){
  pred.knn <- knn(train.X, test.X, train.carnivore, k = i)
  error[i] <- mean(pred.knn != carnivore.test) #Test error (misclassification rate)
}

table(pred.knn, carnivore.test) #Confusion matrix

##          carnivore.test
## pred.knn    0    1
##          0 392  96
##          1  74 160

kvals <- 1:10
plot(1:10, error, xlab = "K", ylab="Error Rate")

```

